

(a)
1) The original statement is

$$\forall x [M(x) \rightarrow \exists y (F(x, y) \wedge H(y))]$$

Negation:

$$\exists x \sim [M(x) \wedge \exists y (F(x, y) \wedge H(y))]$$

$$\exists x [M(x) \wedge \sim \exists y (F(x, y) \wedge H(y))]$$

$$\exists x [M(x) \wedge \forall y (\sim F(x, y) \vee \sim H(y))]$$

$$\exists x [M(x) \wedge \forall y (F(x, y) \rightarrow \sim H(y))]$$

This is equivalent to ~~original~~ negation of original statement
There is someone who is not major and all of his friends don't
need help with this homework.

(b) Original: $\forall x \exists y [R(x, y) \wedge \forall z \sim L(y, z)]$

Negation: $\exists x \forall y [\sim R(x, y) \vee \exists z L(y, z)]$
 $= \exists x \forall y [R(x, y) \rightarrow \exists z L(y, z)]$

There is someone where all roommates likes atleast one person.

(c) ~~Original~~ Original: $\forall x [(x \in A \wedge x \in B) \rightarrow (x \in C \wedge x \in D)]$

Negation: $\exists x \sim [(x \in A \wedge x \in B) \wedge (x \in C \wedge x \in D)]$
 $= \exists x [(x \in A \wedge x \in B) \wedge \sim (x \in C \wedge x \in D)]$

Logically this is negation of original

There is some element x that belongs to $A \cup B$ but don't belong to $C \cap D$

$$= \exists x [(x \in A \wedge x \in B) \wedge (x \notin C \vee x \notin D)]$$

(d) Negation: $\forall x \exists y \sim [(y > x) \vee \exists z (z^2 + 5z = y)]$
 $= \forall x \exists y [y > x \wedge \forall z \sim (z^2 + 5z = y)]$
 $= \forall x \exists y [y > x \wedge \forall z (z^2 + 5z \neq y)]$

2) (a) $\exists x [F(x) \wedge \sim \exists y R(x, y)]$

Negation: $\forall x [\sim F(x) \vee \exists y R(x, y)]$
 $\forall x [F(x) \rightarrow \exists y R(x, y)]$

logically: Everyone in freshman class has atleast one roommate.

(b) $\forall x \exists y L(x, y) \wedge \sim \exists x \forall y L(x, y)$

Negation: $\exists x \forall y \sim L(x, y) \vee \exists x \forall y L(x, y)$

logically: There is someone who likes no one or there is someone who likes everyone.

(c) $\forall a \in A \exists b \in B (a \in C \leftrightarrow b \in C)$

Negation: $\exists a \in A \forall b \in B \sim (a \in C \leftrightarrow b \in C)$

" $\sim (a \in C \leftrightarrow b \in C) \wedge (b \in C \rightarrow a \in C)$

$\exists a \in A \forall b \in B \sim (\sim (a \in C) \vee b \in C) \vee \sim (\sim (b \in C) \vee a \in C)$

$\exists a \in A \forall b \in B (a \in C \wedge \sim b \in C) \vee (b \in C \wedge \sim a \in C)$

logically this is negation

(d) $\forall y > 0 \exists x (ax^2 + bx + c = y)$

Negation: $\exists y > 0 \forall x \sim (ax^2 + bx + c = y)$

(d) We translate this in several steps:

- (i) x is positive and x is a multiple of both y and z and there is no smaller positive number that is a multiple of both y and z .
- (ii) $x > 0 \wedge \exists a(x = ya) \wedge \exists b(x = zb) \wedge \neg \exists w(w > 0 \wedge w < x \wedge (w \text{ is a multiple of both } y \text{ and } z))$.
- (iii) $x > 0 \wedge \exists a(x = ya) \wedge \exists b(x = zb) \wedge \neg \exists w(w > 0 \wedge w < x \wedge \exists c(w = yc) \wedge \exists d(w = zd))$.

2. (a) $\forall x(x + 0 = x)$.

(b) $\forall x \exists y(x + y = 0)$.

(c) $\forall x(x < 0 \rightarrow \neg \exists y(y^2 = x))$.

(d) We translate this gradually:

(i) $\forall x(x > 0 \rightarrow x \text{ has exactly two square roots})$.

(ii) $\forall x(x > 0 \rightarrow \exists y \exists z(y \text{ and } z \text{ are square roots of } x \text{ and } y \neq z \text{ and nothing else is a square root of } x))$.

(iii) $\forall x(x > 0 \rightarrow \exists y \exists z(y^2 = x \wedge z^2 = x \wedge y \neq z \wedge \neg \exists w(w^2 = x \wedge w \neq y \wedge w \neq z)))$.

Exercises

*1. Negate these statements and then reexpress the results as equivalent positive statements. (See Example 2.2.1.)

(a) Everyone who is majoring in math has a friend who needs help with his or her homework.

(b) Everyone has a roommate who dislikes everyone.

(c) $A \cup B \subseteq C \setminus D$.

(d) $\exists x \forall y[y > x \rightarrow \exists z(z^2 + 5z = y)]$.

2. Negate these statements and then reexpress the results as equivalent positive statements. (See Example 2.2.1.)

(a) There is someone in the freshman class who doesn't have a roommate.

(b) Everyone likes someone, but no one likes everyone.

(c) $\forall a \in A \exists b \in B(a \in C \leftrightarrow b \in C)$.

(d) $\forall y > 0 \exists x(ax^2 + bx + c = y)$

3. Are these statements true or false? The universe of discourse is $\mathbb{N}$.

(a) $\forall x(x < 7 \rightarrow \exists a \exists b \exists c(a^2 + b^2 + c^2 = x))$. $\top$

(b) $\exists! x(x^2 + 3 = 4x)$. F (both sq 1 are sol's)

(c) $\exists! x(x^2 = 4x + 5)$. T (sol: 5, -1 also sol's eq's but -1 q/n)

(d) $\exists x \exists y(x^2 = 4x + 5 \wedge y^2 = 4y + 5)$. T

$$= \exists y \exists x (ax^2 + bx + c = y)$$

$$4) \neg \exists x P(x) = \forall x \neg P(x)$$

Replacing $P(x)$ by $\neg P(x)$

$$\neg \exists x \neg P(x) = \forall x \neg \neg P(x)$$

Applying $\neg$ on both sides

$$\exists x \neg \neg P(x) = \neg \forall x \neg P(x) \text{ Proved.}$$

$$5) \neg \exists x \in A P(x)$$

$$= \neg \exists x (x \in A \wedge P(x)) \quad \therefore \text{expanding abbreviation}$$

$$= \forall x \neg (x \in A \wedge P(x)) \quad \therefore \text{Negation}$$

$$= \forall x (\neg x \in A \vee \neg P(x)) \quad \therefore \text{Negation}$$

$$= \forall x (x \in A \rightarrow \neg P(x)) \quad \therefore \neg p \vee q = p \rightarrow q$$

$$= \forall x \in A \neg P(x) \quad \therefore \text{abbreviation}$$

$$6) \exists x (P(x) \vee Q(x)) = \neg \forall x \neg (P(x) \vee Q(x))$$

$$\therefore \text{De Morgan} \quad = \neg \forall x (\neg P(x) \wedge \neg Q(x))$$

$$\therefore \text{distribution of } \neg \quad = \neg (\forall x \neg P(x) \wedge \forall x \neg Q(x))$$

$$\therefore \text{Negation} \quad = \exists x P(x) \vee \exists x Q(x)$$

Proved.

7)

$$\begin{aligned}
 7) & \exists x (P(x) \rightarrow Q(x)) \\
 &= \exists x (\neg P(x) \vee Q(x)) \\
 &= \exists x \neg P(x) \vee \exists x Q(x) \\
 &= \neg \forall x P(x) \vee \exists x Q(x) \\
 &= \forall x P(x) \rightarrow \exists x Q(x)
 \end{aligned}$$

8) Appadim

9) No

$$\begin{aligned}
 10) & \exists x \in A P(x) \vee \exists x \in B P(x) \\
 (a) &= \exists x (x \in A \wedge P(x)) \vee \exists x (x \in B \wedge P(x)) \quad \therefore \text{abb} \\
 &= \exists x [(x \in A \wedge P(x)) \vee (x \in B \wedge P(x))] \quad \therefore \text{Dist} \\
 &= \exists x [(x \in A \vee x \in B) \wedge P(x)] \quad \therefore \text{Dist Law} \\
 &= \exists x \in (A \cup B) P(x) \quad \therefore \text{abb}
 \end{aligned}$$

(b) Prook 1) Prook.

12) We can go either from L.H.S to R.H.S or vice versa.

$$\begin{aligned}
 C \setminus A \subseteq B &: \forall x [(x \in C \wedge x \notin A) \rightarrow x \in B] \\
 &= \forall x [\neg(x \in C \wedge x \notin A) \vee x \in B] \\
 &= \forall x [x \notin C \vee x \in A \vee x \in B] \\
 &= \forall x [x \notin C \vee (x \in A \vee x \in B)] \\
 &= \forall x [x \in C \rightarrow (x \in A \vee x \in B)] \\
 &= \forall x [x \in C \rightarrow x \in (A \cup B)] \\
 &= C \subseteq A \cup B \quad \text{Proved}
 \end{aligned}$$

13) (a) $A \subseteq B$ and $A \cup B = B$

The straight forward & longer way is to simplify $A \cup B = B$
As we know $A = B \iff A \subseteq B \text{ and } B \subseteq A$

So $A \cup B = B$

$$\begin{aligned}
 &= \forall x [(x \in A \vee x \in B) \rightarrow x \in B] \wedge (x \in B \rightarrow (x \in A \vee x \in B)) \\
 &= \forall x [(\cancel{x \in A} \wedge x \in B) \vee x \in B] \wedge (x \in B \vee (x \in A \vee x \in B)) \\
 &= \forall x [(x \in B \vee \cancel{x \in A}) \wedge (x \in B \vee x \in B)] \wedge (x \in B \vee x \in A) \\
 &\quad \therefore \text{Dist, Comm, Associ} \\
 &= \forall x [(x \in A \vee \cancel{x \in A}) \wedge T] \wedge (T \vee x \in A) \quad \therefore \text{Tautology} \\
 &= \forall x [(x \in B \vee \cancel{x \in A}) \wedge T] \quad \therefore \text{Tautology} \therefore \text{Idempotent} \\
 &= \forall x (x \in B \vee x \in B) \quad \therefore \text{Idempotent} \\
 &= \forall x (x \in A \rightarrow x \in B) \\
 &= A \subseteq B \quad \text{Proved.}
 \end{aligned}$$

Other Way. (using formula for exercise 11, 1.5)

$$\begin{aligned}
 &\rightarrow \forall x [(x \in A \vee x \in B) \leftrightarrow x \in B] \\
 &\quad \forall x [x \in A \rightarrow x \in B] \quad \therefore (P \vee Q) \leftrightarrow Q = P \rightarrow Q \\
 &\quad A \subseteq B
 \end{aligned}$$

(b) $A \cap B = A$ ($\therefore$ again we will use ex 11 of sec 1.5)

$$\forall x [(x \in (A \cap B) \rightarrow x \in A) \wedge (x \in A \rightarrow x \in (A \cap B))]$$

$$\forall x [x \in (A \cap B) \leftrightarrow x \in A]$$

$\therefore$ for simplicity, we ~~can~~ open $A \cap B$ braces

$$\begin{aligned}
 & \forall x [(x \in A \wedge x \in B) \leftrightarrow x \in A] \\
 & \forall x [(x \in B \wedge x \in A) \leftrightarrow x \in A] \quad \therefore \text{Comm} \\
 & \forall x [x \in A \rightarrow x \in B] \quad \therefore P \wedge Q \leftrightarrow Q : Q \rightarrow P \\
 & A \subseteq B \text{ Proved}
 \end{aligned}$$

14) Appendix

- *4. Show that the second quantifier negation law, which says that $\neg \forall x P(x)$ is equivalent to $\exists x \neg P(x)$, can be derived from the first, which says that $\neg \exists x P(x)$ is equivalent to $\forall x \neg P(x)$. (Hint: Use the double negation law.)
5. Show that $\neg \exists x \in A P(x)$ is equivalent to $\forall x \in A \neg P(x)$.
- *6. Show that the existential quantifier distributes over disjunction. In other words, show that $\exists x(P(x) \vee Q(x))$ is equivalent to $\exists x P(x) \vee \exists x Q(x)$. (Hint: Use the fact, discussed in this section, that the universal quantifier distributes over conjunction.)
7. Show that $\exists x(P(x) \rightarrow Q(x))$ is equivalent to $\forall x P(x) \rightarrow \exists x Q(x)$.
- *8. Show that $(\forall x \in A P(x)) \wedge (\forall x \in B P(x))$ is equivalent to $\forall x \in (A \cup B) P(x)$. (Hint: Start by writing out the meanings of the bounded quantifiers in terms of unbounded quantifiers.)
9. Is $\forall x(P(x) \vee Q(x))$ equivalent to $\forall x P(x) \vee \forall x Q(x)$? Explain. (Hint: Try assigning meanings to $P(x)$ and $Q(x)$.) *No.*
10. (a) Show that $\exists x \in A P(x) \vee \exists x \in B P(x)$ is equivalent to $\exists x \in (A \cup B) P(x)$.
(b) Is $\exists x \in A P(x) \wedge \exists x \in B P(x)$ equivalent to $\exists x \in (A \cap B) P(x)$? Explain. *Not same because $\exists x \in A$ & $\exists x \in B$ can be different x & not same*
- *11. Show that the statements $A \subseteq B$ and $A \setminus B = \emptyset$ are equivalent by writing each in logical symbols and then showing that the resulting formulas are equivalent.
12. Show that the statements $C \subseteq A \cup B$ and $C \setminus A \subseteq B$ are equivalent by writing each in logical symbols and then showing that the resulting formulas are equivalent.
13. (a) Show that the statements $A \subseteq B$ and $A \cup B = B$ are equivalent by writing each in logical symbols and then showing that the resulting formulas are equivalent. (Hint: You may find exercise 11 from Section 1.5 useful.)
(b) Show that the statements $A \subseteq B$ and $A \cap B = A$ are equivalent.
- *14. Show that the statements $A \cap B = \emptyset$ and $A \setminus B = A$ are equivalent.
15. Let $T(x, y)$ mean " x is a teacher of y ." What do the following statements mean? Under what circumstances would each one be true? Are any of them equivalent to each other?
(a) $\exists y T(x, y)$. *x is teacher of only one student. (can be T)*
(b) $\exists x \exists y T(x, y)$. *there is atleast 1 teacher that is teacher of only 1 student (can be T)*
(c) $\exists x (\exists y T(x, y))$. *x is only teacher of atleast 1 student (can be T)*
(d) $\exists x \exists y (T(x, y) \wedge \neg \exists z T(z, y))$. *there is exactly one student who has exactly/only 1 teacher (can be T)*
(e) $\exists x \exists y (T(x, y) \wedge \neg \exists z T(x, z))$. *exactly one student has only 1 teacher OR exactly 1 teacher has only 1 student*
(f) $\exists x \exists y (T(x, y) \wedge \neg \exists u \exists v (T(u, v) \wedge (u \neq x \vee v \neq y)))$.
there is exactly 1 teacher that teaches exactly 1 student
(Note that it's true if both $u=x$ & $v=y$ & otherwise)

This is about a unique teacher

part (e) and (f) are equivalent only